

AI BUSINESS EXPENSE AND FINANCIAL DOCUMENT ASSISTANT

Finance Operations Automation Project

Prepared by Igbo Chioma Catherine

Executive Summary

This project developed an AI-enabled expense-management system for small-business finance operations. The solution replaces fragmented email, paper receipt and spreadsheet processes with a structured workflow that captures expense information, uses Google Gemini AI to suggest an expense category, stores records in Airtable, applies receipt and amount rules, routes high-value requests to a manager, updates the Lovable dashboard and sends periodic management summaries.

The system retains human control over financial decisions. AI suggests categories and summaries, while explicit business rules and manager decisions determine whether an expense is approved, rejected or sent for clarification. The implemented routing logic uses a NGN 100,000 threshold: submissions without receipts require clarification, qualifying submissions at or below the threshold can be auto-approved, and submissions above the threshold require manager review.

The completed solution demonstrates practical integration of a frontend application, webhooks, AI, database operations, conditional routing, email notifications, APIs and reporting. Testing also identified operational issues such as missing mappings, duplicate records, credit limitations and status inconsistencies, which were reviewed and corrected during implementation.

Contents

1 Project Background and Problem
2 Project Objectives and Scope
3 Tools and System Architecture
4 Data Capture and Database Design
5 AI Categorisation
6 Workflow Implementation
7 Business Rules and Human Oversight
8 Testing and Results
9 Final Outputs and Business Value
10 Challenges Lessons and Future Improvements
11 Conclusion
Appendix Screenshot Placement Checklist

1 Project Background and Problem

Many small organizations manage expenses through disconnected forms, emails, chats, paper receipts and spreadsheets. This makes it difficult to verify supporting documents, track approvals, identify duplicates, monitor expense status and prepare reliable management reports. Manual follow-up also consumes time for employees, managers and finance staff.

The project addresses this problem by centralizing the expense lifecycle. Employees submit requests through a Lovable form, Make receives the data through a webhook, Gemini suggests a category, Airtable stores the record and the workflow applies clear routing rules. Status changes are sent back to the Lovable dashboard through HTTP API calls so that users can view the outcome.

2 Project Objectives and Scope

- Capture complete employee expense information and receipt evidence.

- Store each request in a structured Airtable record with a traceable record identifier.
- Use AI to suggest an appropriate expense category and concise description.
- Apply business rules for missing receipts and the NGN 50,000 approval threshold.
- Route high-value requests to a manager and record the manager decision.
- Notify employees of the relevant outcome and update the frontend dashboard.
- Generate a periodic expense summary for management review.
- Preserve human responsibility for approval, compliance and financial judgement.

Project Boundaries

The solution supports finance operations but does not replace accounting software, professional audit work or financial-policy ownership. It does not allow AI to decide whether an expense is fraudulent, tax-deductible or compliant. Final decisions remain governed by explicit rules and human review.

3 Tools and System Architecture

Tool	Role in the solution
Lovable	Provides the employee submission form and finance dashboard.
Make	Orchestrates the intake, AI, routing, approval, updates and reporting workflows.
Google Gemini AI	Suggests an expense category and produces structured text for workflow use.
Airtable	Stores expense records, statuses, receipt information, manager decisions and comments.
Gmail	Sends manager approval requests and employee or management notifications.
Google Sheets	Captures manager approval-form responses used by the approval scenario.
HTTP API modules	Send updated expense statuses back to the Lovable application.
Webhooks and JSON	Receive frontend submissions and convert AI output into mapped fields.

End to End Architecture

1. An employee submits an expense and receipt information through Lovable.
2. A Make webhook receives the submitted data.
3. Gemini analyzes the purpose and vendor information and suggests a category.
4. The JSON module converts the AI response into structured fields.
5. Airtable creates the expense record.
6. A router evaluates receipt status and the amount threshold.
7. The appropriate route updates Airtable, sends email where necessary and posts the latest status to Lovable.
8. A separate approval scenario processes approved or rejected manager responses.
9. A scheduled reporting scenario aggregates records and emails a management summary.

4 Data Capture and Database Design

The form and database were designed to preserve the original submission while adding workflow-generated information. Core fields include employee name, department, expense date, vendor, amount, purpose, receipt, manager email, AI category, receipt status, approval status, manager decision, manager comments and the Airtable expense record ID.

AI Business Expense & Financial Document Assistant

AI Expense & Financial Document Assistant

Dashboard

Submit Expense

Track Expense

Manager Review

Submit an Expense

Complete the form below to send your business expense for review and reimbursement.

EMPLOYEE DETAILS

Employee Name *

Chioma Igbo

Employee Email *

chioma@company.com

Department *

Select department

Project / Client

e.g. Lagos Retail Rollout

EXPENSE DETAILS

Expense Date *

dd/mm/yyyy

Vendor *

e.g. Chicken Republic

Amount (NGN) *

0.00

Payment Method *

Select payment method

Purpose *

Explain what this expense was for and why it was necessary.

SUPPORTING INFORMATION

Receipt Upload

Choose file

No file chosen

PDF, JPG or PNG up to 5MB

Additional Notes

Anything your manager or finance should know (optional).

Submit Expense

Clear form

Fields marked * are required

Submission checklist

Attach a legible receipt or invoice

Use the exact vendor name on the receipt

Enter amounts in Naira, VAT inclusive

Tag the project or client where applicable

Approval flow

Submitted by employee

Document review & data extraction

Manager approval

Finance reimbursement

Reimbursement window

Approved expenses are paid on the next payroll run. Submissions after the 25th roll into the following month.

Figure 1 Screenshot of Lovable expense submission form

Approval Status	Manager's Email	Manager Comments	Date Submitted	Date Reviewed	AI Category Result	AI Summary Result	Receipt Link	Lovable Expense ID	
Approved	igboco@yahoo.com	Approved to process further...	9/1/2026 2:48pm	1/9/2026 7:31pm	Travel	Return flight with SwiftAir ...	https://c--aa5a9938-7451-...	959615b3-Qaf3-4d81-93ba...	
Approved	igboco@yahoo.com		9/1/2026 2:53pm		Meals	Team lunch at Bistro for th...	https://c--aa5a9938-7451-...	8900419e-6bd4-4a41-99a9...	
Approved	igboco@yahoo.com		9/1/2026 3:41pm		Transportation	Fuel expense of NGN 40,00...	https://c--aa5a9938-7451-...	e7309558-2cc2-4892-96d6...	
Approved	igboco@yahoo.com		9/1/2026 4:19pm		Meals	NGN 45000 for snacks fro...	https://c--aa5a9938-7451-...	6118a92f-de82-4047-b3b6...	
Rejected	igboco@yahoo.com	Double submission	9/1/2026 4:24pm	1/9/2026 7:28pm	Equipment	Bank transfer payment of ...	https://c--aa5a9938-7451-...	6fa0f80f-bcd7-4342-968f-4...	
Needs Clarification	igboco@yahoo.com		9/1/2026 4:30pm		Equipment	Purchase of a wireless keyb...		94cfa3d0-852f-4bc4-a031-...	
Needs Clarification	igboco@yahoo.com		9/1/2026 4:35pm		Training	Diesel expense of NGN 90...		d7e3575b-7290-4a50-9b37...	
Approved	igboco@yahoo.com	Approved to process further	9/1/2026 5:17pm	1/9/2026 7:39pm	Equipment	Purchase of a CPU and offi...	https://c--aa5a9938-7451-...	d1a2f78e-b330-41f7-9d15-...	
Approved	igboco@yahoo.com		9/8/2026 9:38pm		Meals	Team lunch for the AI Auto...	https://c--aa5a9938-7451-...	6ccee269-3f46-4733-9d19-...	
Approved	igboco@yahoo.com	Approved to process further	9/8/2026 9:49pm	8/9/2026 10:01pm	Office Supplies	Bank transfer of NGN 165.0...	https://c--aa5a9938-7451-...	003f6de6-c762-400b-8384...	
Needs Clarification	igboco@yahoo.com		9/12/2026 2:29pm		Equipment	Purchase of a hollow type ...		7cb2d8fd-9ee1-4601-b060...	
Pending	igboco@yahoo.com		9/12/2026 2:29pm		Other	Test expense submission of...	https://c--aa5a9938-7451-...	0c54745f-a744-4ba1-b5df-...	

Expense Record ID	Employee Name	Employee Email	Department	Expense Date	Vendor	Amount (NGN)	Purpose	Receipt	Payment Method	Project/Client	Additional Notes	Receipt Status
1	redc1a48f5a238f2	Chioma Catherine	igboco@aiportal.com	Operations	30/8/2026	SwiftAir Nigeria	\$320,000.00	Return flight for Abuja Tra...	Card	AI Automation Project	Test expense submission	Attached
2	redc1a48f5a238f2	Adedol Vinos	adecolvinos@aiportal.com	Marketing	6/8/2026	Bistro	\$38,500.00	Team lunch during plannin...	Cash	AI Automation Project	Test expense submission	Attached
3	redc1a48f5a238f2	Chioma Catherine Igbo	igboco@aiportal.com	Marketing	2/8/2026	MarsolFuel	\$40,000.00	Fuel for client-site visit	Bank Transfer	AI Automation Project	Test expense submission	Attached
4	redc1a48f5a238f2	Christopher Chibuke	igboco@aiportal.com	Operations	11/8/2026	Tastybite	\$45,000.00	Snacks for business meeting	Bank Transfer	AI Automation Project	Test expense submission	Attached
5	redc1a48f5a238f2	Chioma Catherine Igbo	igboco@aiportal.com	Engineering	2/8/2026	PrimusMart Office	\$200,000.00	Prototype installation	Bank Transfer	AI Automation Project	Test expense submission	Attached
6	redc1a48f5a238f2	Catherine Igbo	igboco@aiportal.com	Procurement	4/8/2026	TechniHub Solution	\$200,000.00	Wireless keyboard and offi...	Card	AI Automation Project	Test expense submission	Not attached
7	redc1a48f5a238f2	Chioma Igbo	igboco@aiportal.com	Marketing	28/8/2026	MarsolGas	\$80,000.00	Diesel for Team Weekend T...	Card	AI Automation Project	Test expense submission	Not attached
8	redc1a48f5a238f2	Chioma Catherine Igbo	igboco@aiportal.com	Operations	4/8/2024	TechniHub Solutions	\$200,000.00	CPU and office monitor	Card	AI Automation Project	Test expense submission	Attached
9	redc1a48f5a238f2	Chioma Igbo	igboco@aiportal.com	Executive Office	2/8/2026	Isaro	\$38,500.00	office team lunch	Cash	AI Automation Project	Test expense submission	Attached
10	redc1a48f5a238f2	Chibuka Christopher	igboco@aiportal.com	Marketing	6/8/2026	AI Automation project	\$165,000.00	Office supply	Bank Transfer	PrimusMart	Test Submission Expense	Attached
11	redc1a48f5a238f2	Adedol Favour	adecolvinos@aiportal.com	Sales	1/8/2026	Halloweem	\$95,000.00	purchase of hollow type pr...	Card	AI Automation Project	Test expense submission	Not attached
12	redc1a48f5a238f2	Chioma Catherine Igbo	igboco@aiportal.com	Operations	5/8/2026	prime mart	\$165,000.00	Test Expense submission	Bank Transfer	AI Automation Project	Test expense submission	Attached

Figure 2 screenshot of the Airtable

5 AI Categorisation

Google Gemini AI reviews descriptive fields such as the department, vendor and purpose. It suggests a category that Make can store in Airtable. The AI output is parsed as JSON so that the category and supporting text can be mapped reliably into later modules.

Categorisation Prompt

You are an expense-categorisation assistant for a business. Use the submitted expense information to suggest a category and concise summary. Review the department, vendor, amount and purpose. Return structured JSON containing the category and summary. Do not approve or reject the expense. If the purpose is unclear, indicate that human clarification is required.

The prompt deliberately separates categorisation from approval. The model assists with interpretation, while Make filters and manager actions control the financial decision.

6 Workflow Implementation

6.1 Expense Intake Categorisation and Routing

The main Make scenario starts when the Lovable application sends data to a custom webhook. Gemini generates the categorisation response, the JSON module parses it, and Airtable creates the record. The router then evaluates three paths: missing receipt, manager approval required and no manager approval required. Each route updates Airtable and posts the resulting status to the Lovable API. The manager-review route also sends an approval email through Gmail.

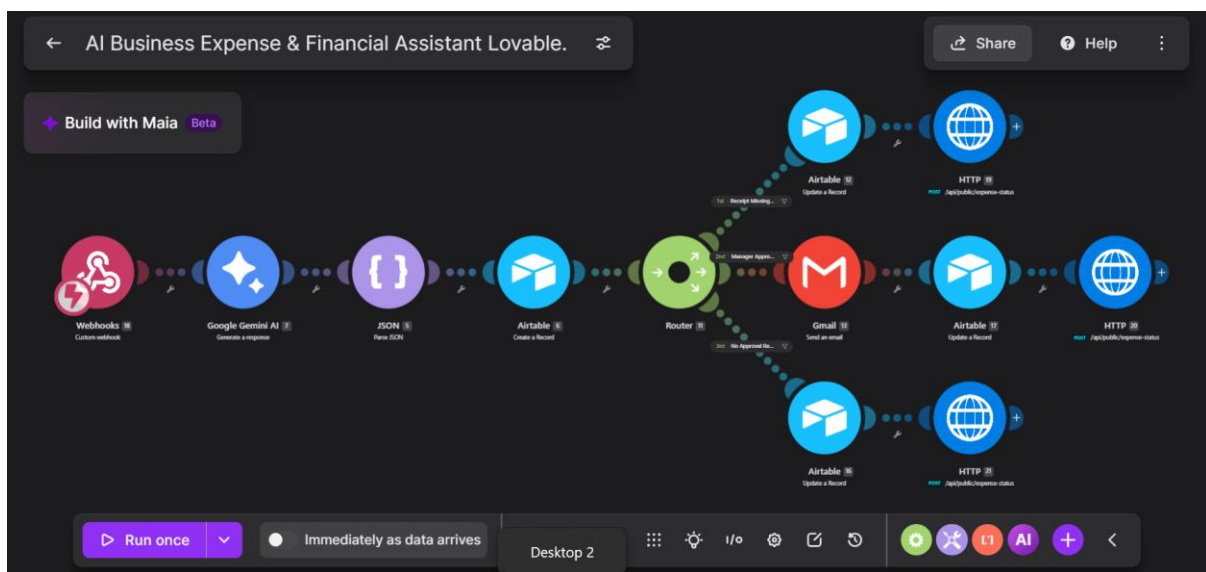


Figure 3. Main Make scenario for expense intake, AI categorisation, Airtable storage, routing and Lovable status updates.

6.2 Manager Approval and Status Update

The approval scenario monitors new manager responses stored in Google Sheets. The router separates approved and rejected decisions. Each route updates the related Airtable record, sends an employee email, retrieves the updated record and posts the final status to the Lovable dashboard through the public expense-status API endpoint.



Figure 4. Manager approval scenario showing approved and rejected processing routes.

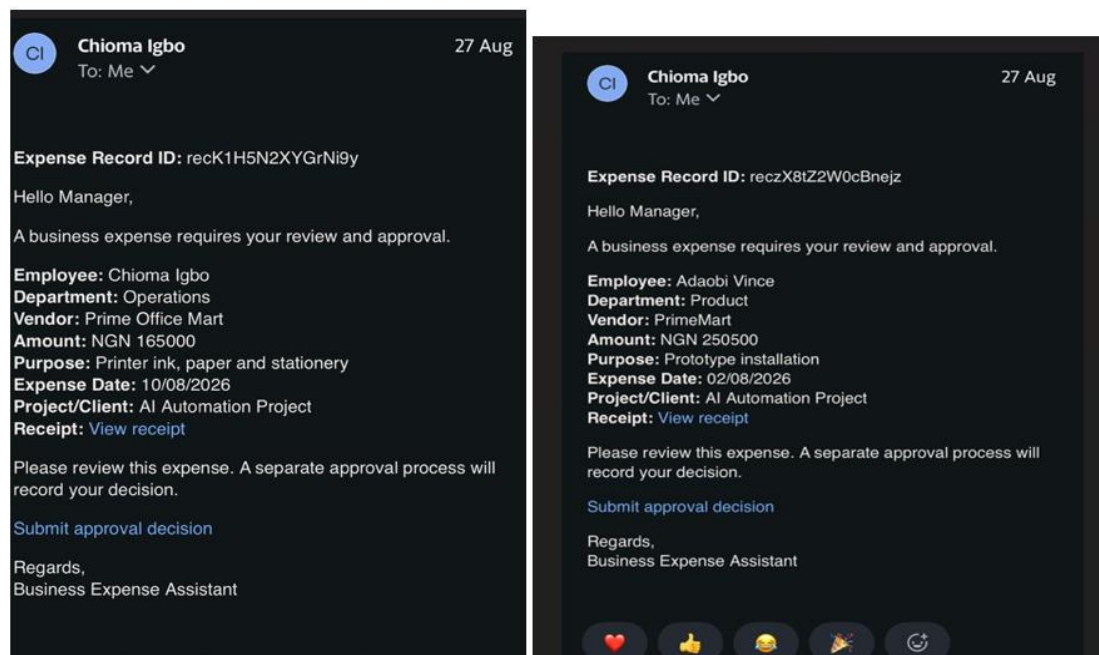


Figure 5 Manager approval request

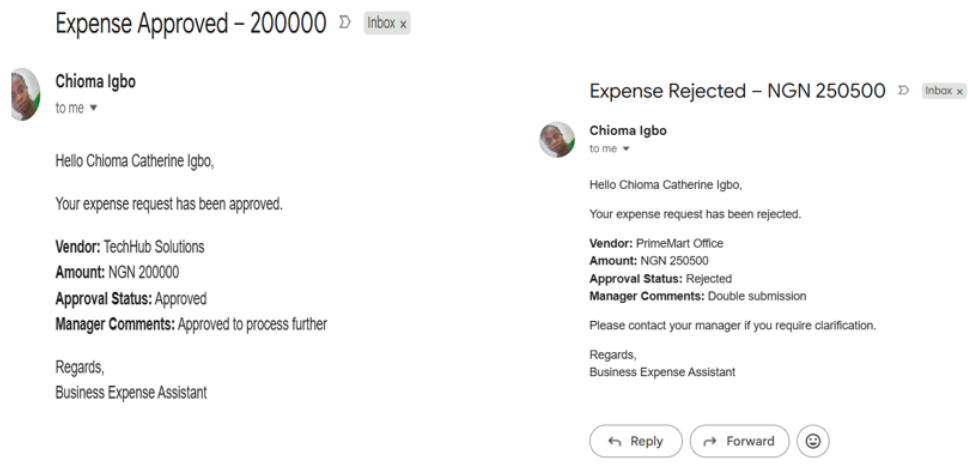


Figure 6 Manager's decision approved and rejected

6.3 Periodic Expense Summary

The reporting scenario searches Airtable records for the selected reporting period, aggregates the record text, sends the combined information to Gemini for a readable management summary and delivers the report through Gmail. The scenario can be scheduled weekly or monthly based on business requirements.

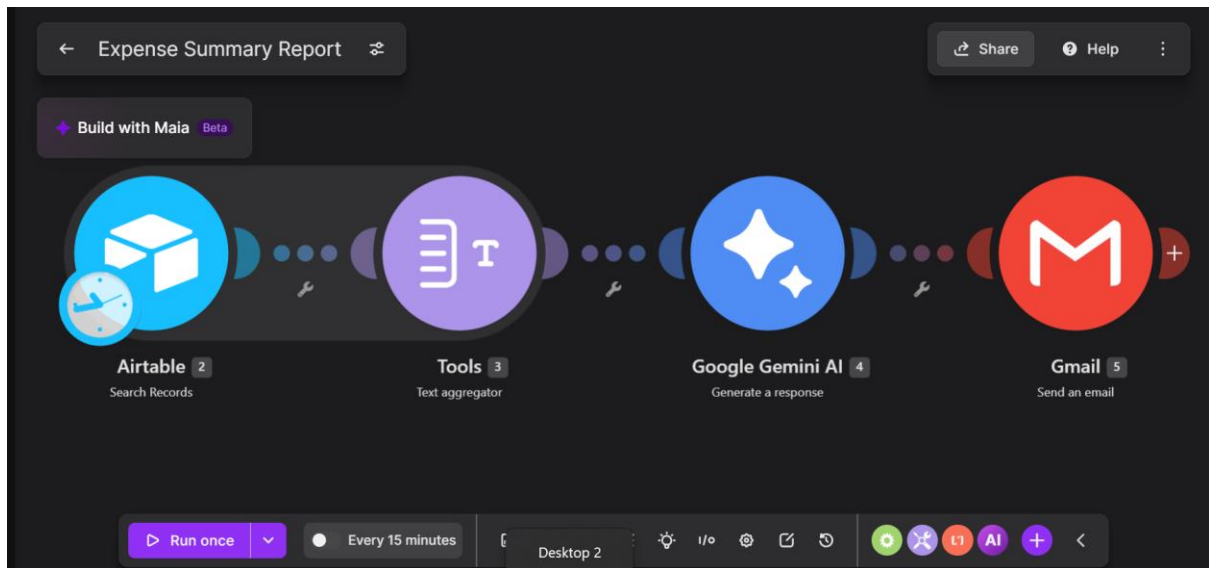


Figure 7. Expense summary scenario using Airtable search, text aggregation, Gemini and Gmail.

WEEKLY BUSINESS EXPENSE SUMMARY

Reporting Period: September 2026

1. Overall Summary

Total Expenses Submitted:	10
Total Amount Submitted:	NGN 1,387,500

2. Status Breakdown

Status	Count	Total Amount (NGN)
Approved	7	NGN 847,000
Pending	0	NGN 0
Rejected	1	NGN 250,500
Needs Clarification	2	NGN 290,000

Status	Count	Total Amount (NGN)
Approved	7	NGN 847,000
Pending	0	NGN 0
Rejected	1	NGN 250,500
Needs Clarification	2	NGN 290,000

3. Expense Totals Grouped by Category

Category	Total Amount (NGN)
Equipment	NGN 650,500
Travel	NGN 320,000
Office Supplies	NGN 165,000
Meals	NGN 122,000

Training	NGN 90,000
Transportation	NGN 40,000

4. Expenses Requiring Clarification / Missing Receipts (2 Records)

Record ID	Employee	Vendor	Amount (NGN)
rec3toBfMDPU9qZrw	Catherine Igbo	TechHub Solution	NGN 200,000
recITalCGxvZuoJ1P	Chioma Igbo	MetroGas	NGN 90,000

5. Duplicate Submissions Analysis

No identical duplicate submission groups (matching exact employee, vendor, amount, category, and receipt link) were identified within this dataset.

6. Management Observation

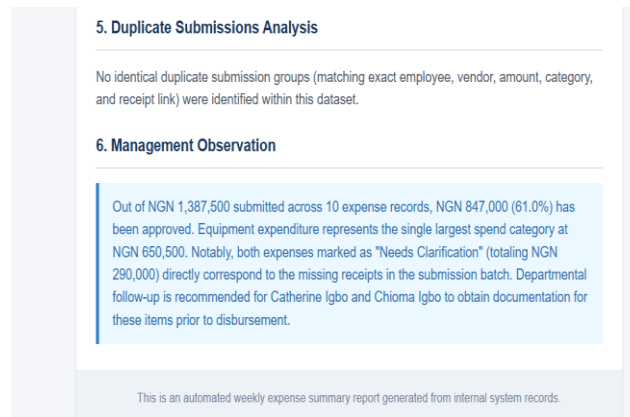


Figure 8 weekly summary report

The Lovable frontend provides a live finance dashboard that displays total submitted value, record count, status totals, recent submissions and spending by department. Records marked Submitted have reached the frontend database but are still awaiting completion of the backend automation or status synchronization

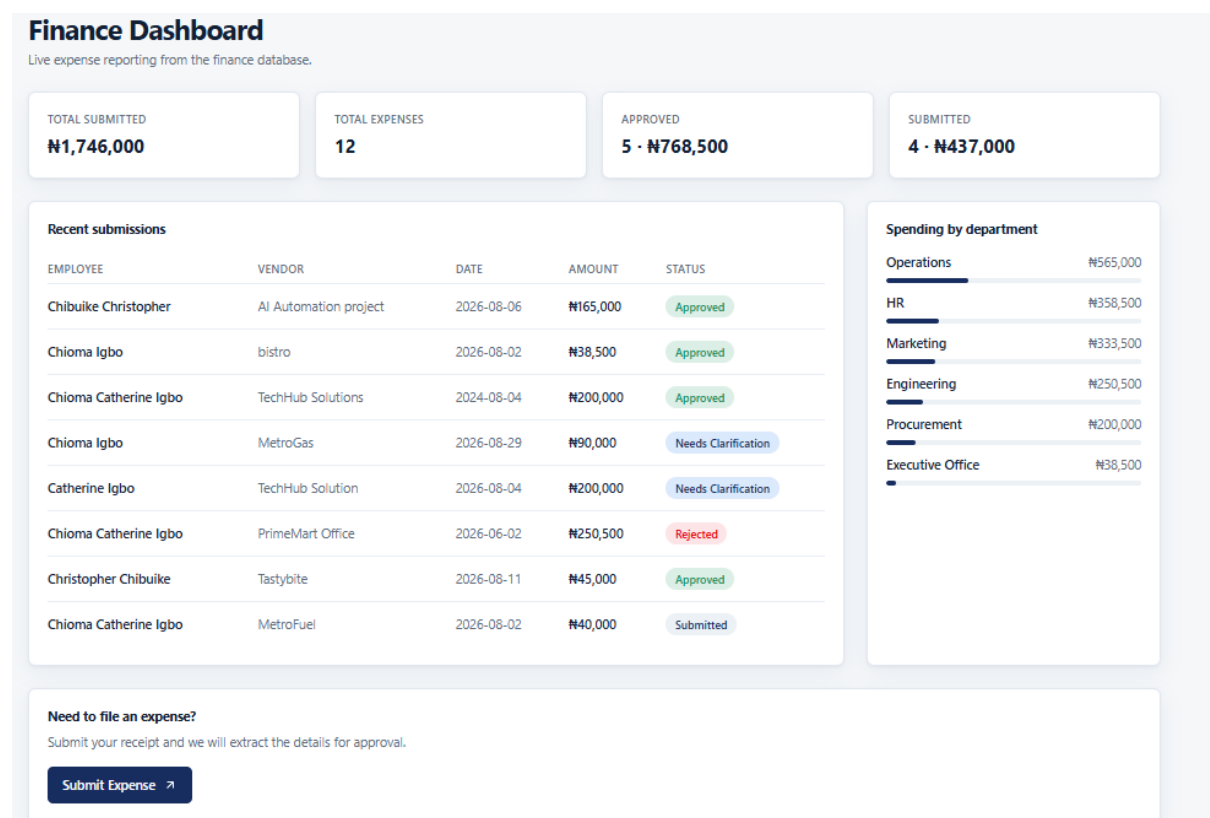


Figure 9 Lovable Dashboard

7 Business Rules and Human Oversight

Condition	Automated action	Human responsibility
Receipt is missing	Set status to Needs Clarification and update the connected systems.	Employee or reviewer supplies or validates the supporting document.
Amount is NGN 50,000 or below and receipt is present	Route to the no-approval path and record the automatic status.	Finance retains the right to review exceptions under company policy.
Amount is above NGN 50,000 and receipt is present	Send an approval request to the manager and mark the record pending.	Manager approves or rejects the request.
Manager approves	Update Airtable, notify the employee and synchronize the dashboard.	Finance proceeds according to payment and control procedures.
Manager rejects	Record rejection, notify the employee and synchronize the dashboard.	Manager provides an appropriate comment or reason.
Possible duplicate or unusual value	Flag for review rather than making an accusation.	A reviewer compares the source records and evidence.

AI does not make the final financial decision because categorization confidence does not prove that a transaction is valid, compliant or properly authorized. Approval requires policy logic, documentary evidence and accountable human judgement.

8 Testing and Results

Testing covered normal submissions, high-value approvals, missing receipts, rejected expenses and duplicate or mapping issues. The table below records the principal scenarios used to validate the workflow. Replace any test description with the exact expense ID if your submission requires record-level evidence.

Test case	Input or condition	Expected outcome	Observed result
1	NGN 38,500 with receipt	Auto-approved because the amount is within the threshold.	Approved during testing.
2	NGN 320,000 with receipt	Sent for manager review, then updated after decision.	Manager approval path completed.
3	NGN 95,000 without receipt	Needs Clarification.	Initially remained Submitted when Make credits were exhausted; used to identify the synchronization dependency.
4	NGN 100,000 manager decision	Rejected route updates record and notifies employee.	Rejected status recorded in the test dataset.
5	Repeated NGN 50,000 submission	Possible duplicate is reviewed rather than automatically rejected.	Duplicate test record identified and corrected during validation.
6	Missing manager-email mapping	High-value request cannot be routed correctly.	Mapping issue identified and corrected in the Lovable form and Make scenario.

Testing Checklist

Area	Validation
Form	Required fields are captured and mapped correctly.
Receipt	Missing receipt follows the clarification route.
AI category	Gemini returns a reasonable and editable category.
Threshold	Amounts above NGN 50,000 require manager approval.
Approval	Approved and rejected responses update the correct Airtable record.
Status synchronization	Airtable and Lovable display the same final status.
Reporting	Summary counts and totals match the selected Airtable records.
Failure handling	Missing mappings, exhausted credits and duplicates are identified and documented.

9 Final Outputs and Business Value

Final Outputs

- A Lovable employee expense-submission form.
- A Make intake and AI-categorisation scenario.
- A structured Airtable expense database.
- Receipt-status and approval-threshold routing.
- A manager approval and rejection workflow.
- Employee and manager email notifications.
- HTTP synchronization with the Lovable finance dashboard.
- A scheduled weekly or monthly management summary.
- Documented test scenarios and exception handling.

Business Value

- Reduces repetitive data entry and manual follow-up.
- Creates a centralized and traceable expense record.
- Improves visibility of pending, approved, rejected and clarification cases.
- Applies the approval threshold consistently.
- Speeds up categorisation while retaining human correction and oversight.
- Supports better management reporting and financial decision-making.
- Provides a scalable foundation for later accounting, reimbursement and policy integrations.

10 Challenges Lessons and Future Improvements

Challenges and Resolutions

Challenge	Resolution or lesson
API and model configuration errors	Validated the API key, selected an available Gemini model and retested the affected module.
Airtable field-type and primary-field issues	Aligned Make mappings with Airtable field types and preserved a reliable record identifier.
Manager email and record-ID mapping	Added the required frontend field and mapped the Airtable record ID through the approval process.
Duplicate NGN 50,000 submission	Reviewed the records and treated duplicate detection as a human-review indicator.
Lovable Submitted status mismatch	Revised status handling so processed records show the correct workflow outcome.
Make credit exhaustion	Recognized that Lovable could accept data while backend processing was unavailable; this demonstrated the need for failure alerts and retry handling.

Future Improvements

- Add automatic receipt data extraction with verification.
- Introduce multi-level approvals by department and amount.
- Add budget-versus-actual reporting and project-level cost tracking.
- Implement stronger duplicate indicators and exception alerts.
- Add retry queues and administrator alerts when Make or an API is unavailable.
- Integrate approved records with accounting or reimbursement software.
- Add role-based access and a formal audit history for production use.

11 Conclusion

The AI Business Expense and Financial Document Assistant demonstrates how AI and automation can improve a practical finance-operations process. The solution connects a user-facing application, AI categorization, database storage, business-rule routing, manager approval, status synchronization

and management reporting. Its most important design principle is that AI assists with classification and summarization, while explicit policy rules and accountable people retain control over financial decisions. The project was built and tested during the Utiva AI Automation Programme.

Appendix Screenshot Placement Checklist

Evidence required	Recommended report location	Status
Expense form	Section 4 Data Capture and Database Design	Figure 1
Airtable base	Section 4 Data Capture and Database Design	Figure 2
Main Make scenario	Section 6.1	Figure 3
Manager approval scenario	Section 6.2	Figure 7
Approval request or decision form	Section 6.2	Figure 6
Employee outcome email	Section 6.2	Figure 5
Summary Make scenario	Section 6.3	Included as Figure 7
Weekly or monthly summary email	Section 6.3	Figure 8
Lovable dashboard	Section 6.3	Figure 9